Separable Essential Ideals and the UBCP Wall

Run `fa_banach_001`, lane 19

June 17, 2026

Source Problem

The source paper is Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257. After proving the commutative unital criterion for $C(K)$, it asks for a noncommutative C^*-algebra version:

> $\{U_\alpha\}_{\alpha\in A}$ of K is called a π-basis (weak basis) for K if every nonempty open subset of K contains some U_α in $\mathcal{U}$ [37, Page 15].
>
> In the setting of commutative C^*-algebra, [25, Theorem 2.1] can be reformulated as follows. Note that [25, Theorem 2.1] only deals with real continuous functions. The complex case is a direct consequence of the real case.
>
> **Corollary 3.9.** *Let $\mathcal{A}$ be a unital commutative C^*-algebra. Then the followings are equivalent:*
>
> (1) *$\mathcal{A}$ has the BCP;*
> (2) *the pure state space $\mathfrak{P}(\mathcal{A})$ has a countable π-basis;*
> (3) *$\mathcal{A}$ has the UBCP.*
>
> In particular, there exist commutative C^*-algebras containing no minimal projections but having the BCP. For example, the uniform norm closure of the linear span of Haar functions [24, p.150].
>
> We conclude this section by the following question.
>
> **Question 3.** *It will be interesting to find a non-commutative version of Corollary 3.9, that is, how to characterize those C^*-algebras $\mathcal{A}$ having BCP/UBCP.*

Definitions

Let X be a normed space. It has BCP if S_X is covered by countably many open balls not containing the origin. Equivalently, there are countably many centres x_n such that

$$S_X \subseteq \bigcup_n B(x_n, \|x_n\|).$$

It has UBCP if there is such a cover with bounded radii and with all balls uniformly separated from a neighbourhood of the origin.

The earlier packet

`2311.14257_bcp_countable_capture_exact_equivalence`

proved that a nonzero normed space X fails BCP if and only if it has countable capture: for every countable $C \subset X$ there is $u \in S_X$ such that

$$\operatorname{dist}(c, \mathbb{F}u) = \|c\| \qquad (c \in C).$$

Main Positive Theorem

Theorem 1 (separable essential ideals force UBCP)**.** *Let A be a C^*-algebra. If A contains a separable essential closed two-sided ideal I, then A has UBCP.*

Proof. Since I is essential, the canonical map from A into the multiplier algebra $\mathcal{M}(I)$ is isometric. Thus, for $a \in A$,

$$\|a\| = \sup\{\|ax\| : x \in I,\ \|x\| \leq 1\}.$$

Equivalently, if (e_α) is a positive contractive approximate identity for I, then $\|ae_\alpha\| \to \|a\|$. Indeed, choose $x \in I$ with $\|x\| \leq 1$ and $\|ax\|$ close to $\|a\|$, and then use $e_\alpha x \to x$.

Let D be a countable dense subset of the closed unit ball of I, and put

$$D_0 = \{d \in D : \|d\| > 13/16\}.$$

For every $d \in D_0$, take the centre $2d$ and the fixed radius $9/8$. This is a countable family of balls. Its radii are bounded, and every centre has norm larger than $13/8$, so all balls avoid $B(0, 1/2)$.

We show that these balls cover S_A. Fix $a \in S_A$. By the first paragraph, choose a positive contraction $e \in I$ such that

$$\|ae\| > 7/8.$$

Since $ae \in I$ and $\|ae\| \leq 1$, choose $d \in D$ with

$$\|d - ae\| < 1/16.$$

Then $\|d\| > 7/8 - 1/16 = 13/16$, so $d \in D_0$.

In the unitization of $\mathcal{M}(I)$, the element $1-2e$ is a contraction, because e is a positive contraction. Hence

$$\begin{aligned}\|a - 2d\| &\leq \|a - 2ae\| + 2\|ae - d\| \\ &= \|a(1-2e)\| + 2\|ae - d\| \\ &< 1 + 1/8 = 9/8.\end{aligned}$$

Thus $a \in B(2d, 9/8)$. The unit sphere is covered by the countable family above, with uniformly bounded radii and a uniform gap from the origin. Therefore A has UBCP. □

Corollary 1. *If a C^*-algebra has BCP but not UBCP, then it has no separable essential ideal.*

Proof. This is the contrapositive of Theorem 1, using UBCP $\Rightarrow$ BCP. □

Examples Covered

The theorem recovers several positive regions by a single argument. Every separable C^*-algebra has itself as a separable essential ideal. If H is separable, then $\mathcal{K}(H)$ is a separable essential ideal in $\mathcal{B}(H)$, so $\mathcal{B}(H)$ has UBCP. If (A_n) is a sequence of separable C^*-algebras, then

$$\bigoplus_{n=1}^{\infty} A_n$$

is a separable essential ideal in $\prod_n A_n$, so the countable product has UBCP. This gives a short common explanation for the countable-reservoir positive cases in the lane.

Quantitative UBCP Reformulation

Lemma 1 (bounded anti-capture form of UBCP)**.** *For a normed space X, the following are equivalent:*

1. *X has UBCP;*
2. *there are a bounded countable set $D \subset X$ and a number $\delta > 0$ such that for every $u \in S_X$ some $d \in D$ satisfies*
$$\|u - d\| < \|d\| - \delta.$$

Consequently, if X fails UBCP, then for every bounded countable $D \subset X$ and every $\delta > 0$ there is $u \in S_X$ such that
$$\|u - d\| \geq \|d\| - \delta \qquad (d \in D).$$

Proof. If (2) holds, the balls $B(d, \|d\| - \delta)$, $d \in D$, cover S_X. Their radii are bounded because D is bounded, and they all avoid $B(0, \delta)$. Hence X has UBCP.

Conversely, a UBCP cover has bounded radii and avoids some ball $B(0, \delta)$. If its centres are d_n and its radii are r_n, then
$$r_n \leq \|d_n\| - \delta$$
for every n. Since S_X is covered and the radii are bounded, the centres are bounded. This gives (2), after replacing each r_n by $\|d_n\| - \delta$ if necessary.

The final sentence is the negation of (2). □

The Remaining Wall: Asymptotic Capture

Theorem 2 (BCP but not UBCP gives quotient capture)**.** *Let X be a nonzero normed space over $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. Assume X has BCP but not UBCP. Let $C = \{c_1, c_2, \ldots\} \subset X$ be any countable set witnessing the positive side of the exact BCP/capture equivalence, i.e. for every $u \in S_X$ there is $c \in C$ with*
$$\operatorname{dist}(c, \mathbb{F}u) < \|c\|.$$
Then the sequence quotient
$$X_\infty = \ell_\infty(X)/c_0(X)$$
contains $v \in S_{X_\infty}$ such that, for the diagonal embedding $\Delta : X \to X_\infty$,
$$\operatorname{dist}(\Delta c, \mathbb{F}v) = \|c\| \qquad (c \in C).$$

Proof. Let $Q_\mathbb{F} = \mathbb{Q}$ in the real case and $Q_\mathbb{F} = \mathbb{Q} + i\mathbb{Q}$ in the complex case. For $m \geq 1$, put
$$D_m = \{qc_j : 1 \leq j \leq m,\ q \in Q_\mathbb{F},\ |q| \leq m\}.$$
This is a bounded countable subset of X. Since X does not have UBCP, Lemma 1 gives $u_m \in S_X$ such that
$$\|u_m - y\| \geq \|y\| - 1/m \qquad (y \in D_m).$$
Let v be the class of the bounded sequence (u_m) in $\ell_\infty(X)/c_0(X)$. Its quotient norm is $\limsup_m \|u_m\| = 1$, so $v \in S_{X_\infty}$.

Fix $c = c_j \in C$ and $\lambda \in \mathbb{F}$. If $\lambda = 0$, then $\|\Delta c - \lambda v\| = \|c\|$. Assume $\lambda \neq 0$, and put $\mu = \lambda^{-1}$. Let $\varepsilon > 0$. Choose $q \in Q_\mathbb{F}$ with
$$|q - \mu| < \varepsilon.$$

For all sufficiently large m, we have $m \geq j$ and $|q| \leq m$, hence $qc \in D_m$. Therefore

$$\begin{aligned}\|u_m - \mu c\| &\geq \|u_m - qc\| - \|(q-\mu)c\| \\ &\geq |q|\,\|c\| - 1/m - \varepsilon\|c\| \\ &\geq (|\mu| - 2\varepsilon)\|c\| - 1/m.\end{aligned}$$

Taking the limsup and then letting $\varepsilon \downarrow 0$ gives

$$\|v - \mu\Delta c\| \geq |\mu|\,\|c\|.$$

Multiplying by $|\lambda|$, we obtain

$$\|\Delta c - \lambda v\| \geq \|c\|.$$

Since this holds for every scalar λ, and equality is obtained at $\lambda = 0$, the desired distance equality follows. □

Corollary 2 (shape of a C*-counterexample)**.** *If a C^*-algebra A has BCP but not UBCP, then:*

1. *A has no separable essential ideal;*
2. *for every countable BCP witness $C \subset A$, the quotient $\ell_\infty(A)/c_0(A)$ contains a unit vector that captures the diagonal copy of C.*

Proof. The first assertion is Corollary 1; the second is Theorem 2. □

What Remains

Theorem 1 is a substantial positive theorem, but it is not a full solution of Question 3. It reduces the possible failure of

$$\text{BCP} \Rightarrow \text{UBCP}$$

for C^*-algebras to algebras with no separable essential ideal. The most direct remaining route is therefore:

prove that every C^*-algebra with BCP contains a separable essential ideal.

If that statement is true, then Theorem 1 immediately gives the full implication BCP $\Rightarrow$ UBCP for C^*-algebras. If it is false, Corollary 2 says any counterexample must hide the countable BCP witness in A while allowing exact line capture of that witness in the sequence quotient.

Novelty and Search Bounds

Local run indexes were searched for arXiv:2311.14257, UBCP, SBCP, essential ideal, separable essential ideal, multiplier algebra, countable capture, and the previous lane-19 C*-BCP packets. The earlier essential-Fell packet proves a type-I local chart theorem; it does not contain the general separable-essential-ideal compression proof above. The Liu–Liu–Lu–Zheng paper arXiv:2111.04921 was checked for the definitions and the vector-valued $C_0(K, X)$ UBCP stability theorem. No exact version of Theorem 1 or Theorem 2 was found in the local packet index.

Human Review Focus

Reviewers should check:

1. the use of essentiality to view A isometrically inside $\mathcal{M}(I)$;
2. the passage from arbitrary contractions in I to positive contractions e using an approximate identity;
3. the fixed estimate $\|1 - 2e\| \leq 1$;
4. the quantitative negation of UBCP in Lemma 1;
5. the rational-scalar diagonalization in Theorem 2.

References

[1] Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257, 2023.

[2] Minzeng Liu, Rui Liu, Jimeng Lu, and Bentuo Zheng, *Ball covering property from commutative function spaces to non-commutative spaces of operators*, arXiv:2111.04921, 2021.

[3] Bruce Blackadar, *Operator Algebras: Theory of C^*-Algebras and Von Neumann Algebras*, Encyclopaedia of Mathematical Sciences 122, Springer, 2006.